

Causal Inference - Stage 2

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Introduction to Causal Inference 097400

1 Causal Question

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What is the causal effect of the implementation of the Manhattan congestion pricing program on the average speed of Central Business District (CBD)-exposed Metropolitan Transit Authority (MTA) bus routes, relative to the change in speed of comparable non-CBD-exposed routes?

2 Existing Knowledge

The following evidence guides our causal question:

Evaluating the Traffic and Emissions Impacts of Congestion Pricing in New York City

Because this study predates the January 2025 implementation of Manhattan’s congestion pricing by five years, it relies on predictive modeling to simulate the impacts of various tolling tiers. The authors predict that tolling will reduce private vehicle volume within the CBD as travelers switch transit modes, carpool, adjust departure times, reroute, or cancel discretionary trips. This ex-ante modeling supports the theoretical mechanism for an increase in bus speeds. Crucially, the study notes that congestion relief will not be uniform across all corridors; factors such as pedestrian density, delivery vehicles, and intersection geometry will continue to cause delays on specific streets.

The Short-Run Effects of Congestion Pricing in New York City

This paper evaluates the empirical short-run effects of the Central Business District Tolling Program using data from the first three months following its launch on January 5, 2025. The authors establish that the policy successfully increased traffic speeds within the CBD and across peripheral corridors of the metropolitan network. To isolate these effects, their empirical design compares New York City’s traffic patterns against a control group of large metropolitan areas that did not implement tolling—including Boston. This framework provides a direct precedent for our choice of control group and suggests that leveraging a generalized synthetic control (GSC) approach could offer a robust strategy for constructing a valid counterfactual.

More Bus Lanes Are Coming To The City’s Busiest Bus Route

This industry report outlines the operational landscape of high-density bus corridors immediately prior to tolling implementation. It highlights the strategic expansion of dedicated bus lanes and Automated Camera Enforcement (ACE) deployed by the city to mitigate transit demand spikes and position public transit as a viable alternative to driving. From an econometric perspective, these targeted transit interventions represent critical time-varying confounders. Because simultaneous policies like ACE and newly painted bus lanes independently affect transit speeds, our empirical model should try to control for these route-specific priority measures to isolate the true causal impact of congestion pricing.

3 Data

NYC

We intend to use several datasets from New York State Open Data and the Metropolitan Transportation Authority (MTA). Together, these datasets allow us to identify which bus routes are exposed to Manhattan’s Central Business District (CBD), measure bus speeds before and after the implementation of congestion pricing, and check whether route-level exposure classifications remain valid over time.

The main outcome dataset will be *MTA Bus Speeds: Beginning 2015*. This dataset reports monthly bus performance by route, borough, day type, trip type, and service period. It includes total mileage, total operating time, and average speed for each observation. We will use this dataset to construct the main route-level panel.

The treatment assignment will come primarily from the *MTA Central Business District Bus Routes* dataset. This dataset identifies MTA bus routes that intersect with the CBD and describes whether each route crosses the CBD or is entirely contained within it. A limitation of this dataset is that it was last updated in July 2024 (before the post-treatment period). To overcome this, we may need to use the *MTA Bus Routes* and *MTA Central Business District Bus Speeds: Beginning 2023* datasets, which are both updated to the present time, as validation for the treatment. If we find inconsistencies, the *MTA Bus Routes* dataset can be used in conjunction with the *MTA Central Business District Geofence* dataset to manually sort treated routes through their route geometries.

The resulting main analysis dataset will be a route-month-period panel containing route ID, month, average speed, treatment status, and a post-treatment indicator.

Boston

As an external comparison, we will use Massachusetts Bay Transit Authority (MBTA) bus arrival and departure data from Boston, where no congestion pricing policy was implemented. The dataset contains trip-level observations by service date, route, direction, half-trip, stop, and timepoint. Since it does not directly report route-level average speed, we will construct a this measure using observed arrival and departure times along with distance traveled.

4 Causal Assumptions

Since we are structuring this as a Difference-in-Differences (DiD) design, we must assume Consistency, Parallel Trends, and No Pretreatment Effect.

Consistency. This is fairly straightforward, the treated units are clearly defined by a physical barrier, and for each observation, the observed average speed equals the potential outcome under the route’s actual exposure status.

Parallel Trends. We assume that, absent congestion pricing, CBD-exposed and non-exposed routes would have followed parallel trends in average bus speed. I.e., any post-policy divergence should be attributed to congestion pricing, rather than pre-existing differences in their speed trajectories. From a preliminary analysis we see this assumption is reasonable:

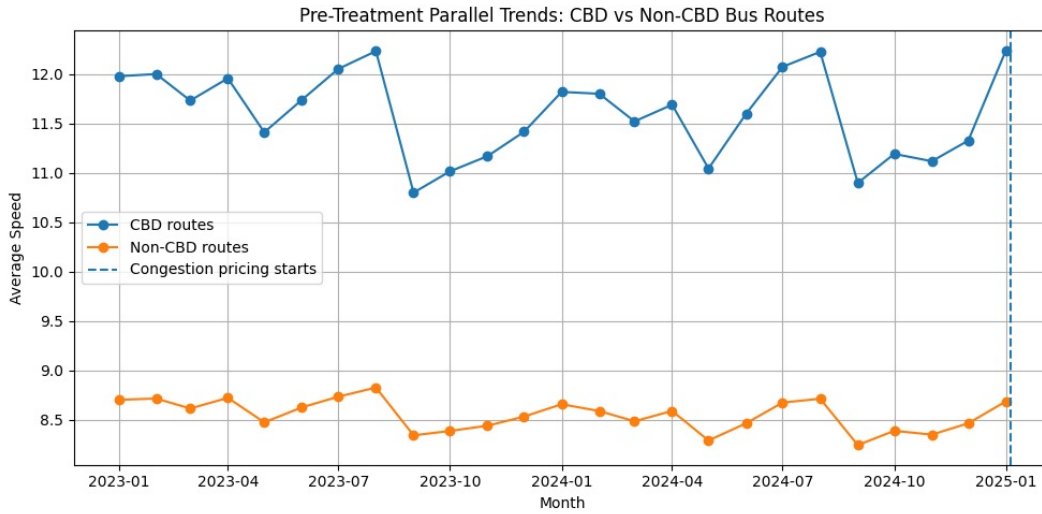
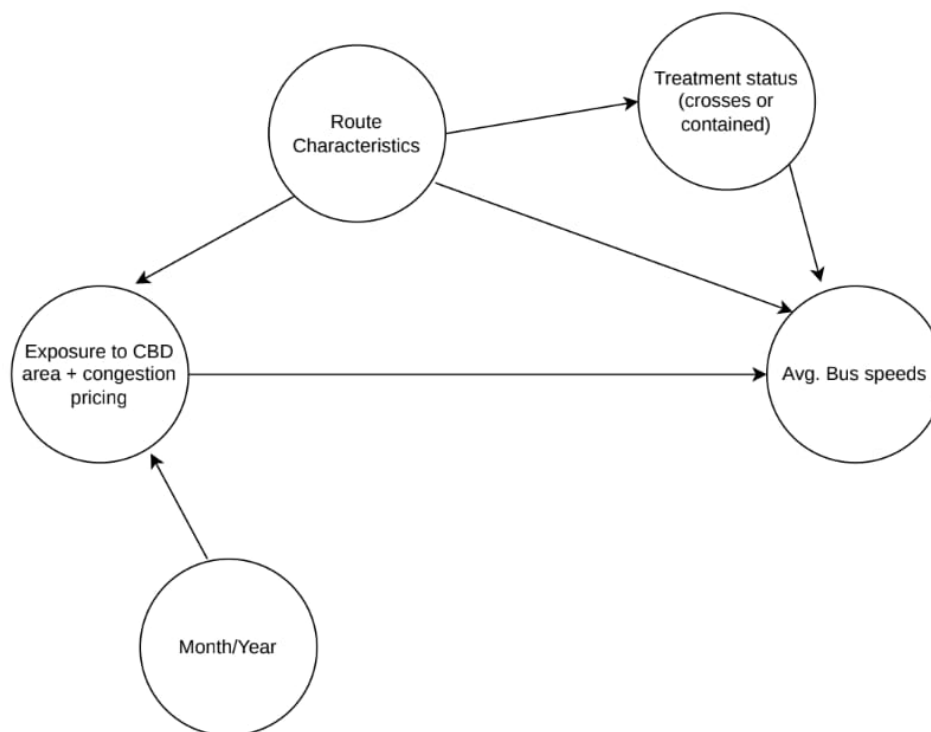


Figure 1: Pre-treatment parallel trends in average bus speeds for CBD and non-CBD routes

No Pretreatment Effect. We assume that density did not change significantly before the implementation of congestion pricing. Although there is a spike in average bus speed in the months leading to congestion pricing, it seems to be within the range of previous averages and so we assume no pre-treatment effect.

5 Pearl's causal graph



6 Possible Challenges

One challenge within the data is "strength" of treatment among the treated routes. Even though we can differentiate between intersecting routes and routes contained in the CBD, it is hard to differentiate between routes which barely intersect and routes which are mostly in the CBD. This may blur the treatment effect.

In addition, some transportation design changes might have occurred in conjunction with the policy to make it more effective. This could include bus lane construction or increased bus lane camera enforcement (making sure cars are not blocking the lanes). This, too, might blur the treatment effect.

Lastly, although the MBTA data is used as a robustness check, it is not structured in the same way as the MTA data. Specifically, it does not have average speeds in the data and must be constructed, which may lead to additional error.

7 Estimation methods

As previously stated, we are setting this up as a difference-in-differences estimation. This method is appropriate because congestion pricing was implemented at a specific point in time, creating a natural experiment where certain bus routes were exposed to the CBD while others were not. The

DiD estimand is the average treatment effect on the treated routes (ATT): the effect of congestion pricing exposure on the bus routes that were actually exposed to the CBD.

The baseline regression estimation is as follows:

$$Y_{rmp} = \alpha_r + \lambda_m + \delta_p + \beta(\text{CBDExposed}_r \times \text{Post}_m) + \varepsilon_{rmp}$$

where (Y_{rmp}) is the average speed of route (r) in month (m) and time period (p). (α_r) are route fixed effects, which absorb time-invariant differences across routes. (λ_m) are month fixed effects, which absorb time shocks common to all routes. (δ_p) controls for systematic differences between time periods, such as daytime and nighttime. The coefficient (β) is the DiD estimate of the effect of CBD exposure following the implementation of congestion pricing.

If the parallel trends assumption is violated or the regular DiD framework fails to produce a credible comparison group, we will employ a synthetic control method. This approach allows us to construct a weighted combination of control routes whose pre-treatment speed trends more closely match those of the treated, CBD-exposed routes.

8 Sensitivity Checks

We propose two primary sensitivity checks to validate our main findings. The first evaluates whether the congestion pricing policy itself drives the treatment effects, while the second assesses the sensitivity of the treatment definition.

To confirm that congestion pricing is the primary driver of the observed effects—rather than confounding factors such as new bus lane construction or enhanced enforcement—we will conduct several robustness checks. These include utilizing alternative control groups (such as routes in Boston) and applying the aforementioned synthetic control method. To test the sensitivity of the treatment definition, we will restrict the treated sample to routes operating entirely within the CBD, testing whether the effect size increases with greater geographic exposure.

Finally, we will bootstrap the ATT estimates by resampling at the route level to construct robust confidence intervals.